

Chenxu Zhao

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Education

Iowa State University

Ph.D. in Computer Science

Ames, IA

Jan. 2023 – Jun. 2027 (Expected)

Iowa State University

M.S. in Computer Science

Ames, IA

Jan. 2023 – May 2025

The Chinese University of Hong Kong, Shenzhen

B.S. in Statistics

Shenzhen, China

Aug. 2018 – May 2022

Internship Experience

Amazon

Bellevue, WA

Applied Scientist Intern, Agent Evaluation & Self-Evolving Agents

Jun. 2026 – Sep. 2026

- Researched and tailored evaluation systems for complex AI agents to support dedicated self-improvement loops, focusing on multi-metric design, robustness enhancement, targeted feedback refinement, and reward-hacking prevention.
- Designed an autonomous self-improvement framework that enables AI agents to iteratively optimize their behavior using established evaluation signals and feedback.

Selected Research Projects

Large Language Model Unlearning Security Vulnerabilities

- Built **LUSB**, a security benchmark for LLM unlearning that formalizes attacks across four threat scenarios and evaluates **16** attack/defense methods, **13** LLM architectures (up to 70B), **12** datasets, and **9** unlearning algorithms.
- Designed subset-optimization attacks that achieved up to **3×** improvement over baseline attacks.

Vision-Language Model Uncertainty Quantification

- Developed a reasoning-answer uncertainty framework with **finite-sample guarantees** and a **Shapley-value-based** module for attributing uncertainty to data and reasoning steps.
- Validated on **3 VLM architectures** across challenging multimodal benchmarks; published at **ACL 2026**.

Large Language Model Unlearning Privacy Vulnerabilities

- Constructed **PrivUB**, benchmarking **21** attacks/defenses across **11** datasets, **10** models, and **10** unlearning algorithms for membership inference, reconstruction, and knowledge leakage.
- Showed that perturbing unlearned data increased recovered-knowledge ROUGE by **79%** on Llama-3-8B/RWKU with RMU; published at **AAAI 2026 (Oral)**.

Membership Inference Attacks with Reliable Control

- Proposed **MIAFdR**, a conformal-inference wrapper that provides **provable false discovery rate control** for existing membership inference attacks.
- Validated across **3** datasets, **4** architectures, and **5+** MIA families with $< 0.01\%$ runtime overhead; published at **ICCV 2025**.

Selected Publications [Google Scholar]

Towards Benchmarking Privacy Vulnerabilities in Selective Forgetting with LLMs. Wei Qian, **Chenxu Zhao**, Yangyi Li, Mengdi Huai. *AAAI 2026, Oral*.

Membership Inference Attacks with False Discovery Rate Control. **Chenxu Zhao**, Wei Qian, Aobo Chen, Mengdi Huai. *ICCV 2025*.

Rethinking Adversarial Robustness in the Context of the Right to be Forgotten. **Chenxu Zhao**, Wei Qian, Yangyi Li, Aobo Chen, Mengdi Huai. *ICML 2024*.

Static and Sequential Malicious Attacks in the Context of Selective Forgetting. **Chenxu Zhao**, Wei Qian, Rex Ying, Mengdi Huai. *NeurIPS 2023*.